MEDIUM

ADVANCED MATH

Advanced Math

07

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

$$fx = x^2 - 18x - 360$$

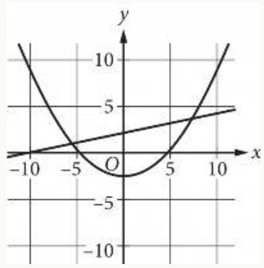
If the given function f is graphed in the xy-plane, where $y = fx$, what is an x-intercept of the graph?

A. $-12, 0$

B. $-30, 0$

C. $-360, 0$

D. $12, 0$



A system of equations consists of a quadratic equation and a linear equation. The equations in this system are graphed in the xy-plane above. How many solutions does this system have?

A. 0

B. 1

C. 2

D. 3

Q3 ADVANCED MATH

Which expression is equivalent to $7x^3 + 7x - 6x^3 - 3x$?

A. $x^3 + 10x$

B. $-13x^3 + 10x$

C. $-13x^3 + 4x$

D. $x^3 + 4x$

x	y
1	11
2	19
3	a

The table shows three values of x and their corresponding values of y for the equation $y = 42^x + 3$. In the table, a is a constant. What is the value of a?

A. 67

B. 35

C. 32

D. 27

Q5 ADVANCED MATH

$$P = N19 - C$$

The given equation relates the positive numbers P, N, and C. Which equation correctly expresses C in terms of P and N?

A. $C = \frac{19 + P}{N}$

B. $C = \frac{19 - P}{N}$

C. $C = 19 + \frac{P}{N}$

D. $C = 19 - \frac{P}{N}$

Q6

ADVANCED MATH

$$\sqrt[3]{x^3y^6}$$

Which of the following expressions is equivalent to the expression above?

A. y^2

B. xy^2

C. y^3

D. xy^3

Q7 ADVANCED MATH

The area of a triangle is 270 square centimeters. The length of the base of the triangle is 12 centimeters greater than the height of the triangle. What is the height, in centimeters, of the triangle?

A. 15

B. 18

C. 30

D. 36

Q8

GRID-IN ADVANCED MATH

$$x^2 - 5x - 24 = 0$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ADVANCED MATH

Which expression is equivalent to $8x^3 + 8 - x^3 - 2$?

A. $8x^3 + 6$

B. $7x^3 + 10$

C. $8x^3 + 10$

D. $7x^3 + 6$

Q10

ADVANCED MATH

The function $f(w) = 6w^2$ gives the area of a rectangle, in square feet (ft^2), if its width is w ft and its length is 6 times its width. Which of the following is the best interpretation of $f(14) = 1{,}176$?

A. If the width of the rectangle is 14 ft, then the area of the rectangle is $1{,}176$ ft^2.

B. If the width of the rectangle is 14 ft, then the length of the rectangle is $1{,}176$ ft.

C. If the width of the rectangle is $1{,}176$ ft, then the length of the rectangle is 14 ft.

D. If the width of the rectangle is $1{,}176$ ft, then the area of the rectangle is 14 ft^2.